

Germanium

Ge (Ge I). Strongest classified transitions below 2 μm (NIST): 20 levels, 54 lines.

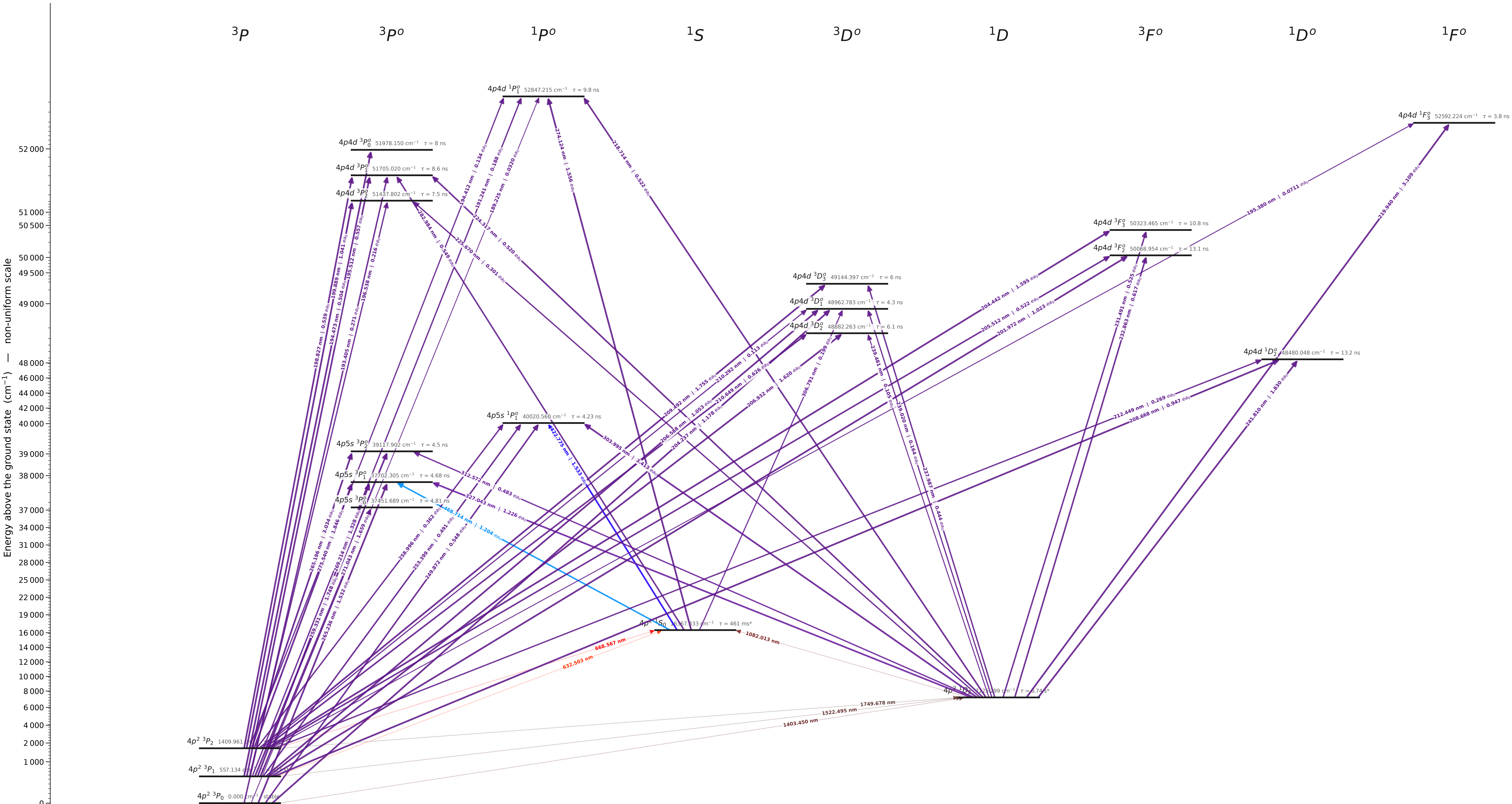
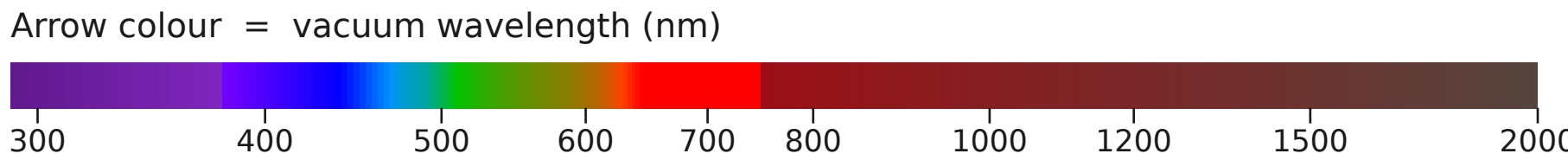
Ge⁺ ionisation limit 63713.24 cm⁻¹ = 7.89943 eV (156.953 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{e}r \mathbf{J}'\rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4p^2\ ^3P_1 - 4p5s\ ^3P_1^o$	269.2140	269.1341	1113.5842	1.328	6.100e+07	34.01	22.1	269.13 nm line used for collinear laser spectroscopy of G...
$4p^2\ ^3P_1 - 4p^2\ ^1S_0$	632.5031	632.3282	473.9778	-	1.000e+00	3.452e-07	46.1	forbidden line within the ground configuration
$4p^2\ ^3P_2 - 4p^2\ ^1S_0$	668.5666	668.3821	448.4107	-	6.800e-02	3.452e-07	3.14	forbidden line within the ground configuration
$4p^2\ ^1D_2 - 4p^2\ ^1S_0$	1082.0129	1081.7165	277.0692	-	1.100e+00	3.452e-07	50.7	forbidden line within the ground configuration
$4p^2\ ^3P_0 - 4p^2\ ^1D_2$	1403.4499	1403.0663	213.6111	-	3.000e-05	2.361e-08	0.0202	forbidden line within the ground configuration
$4p^2\ ^3P_1 - 4p^2\ ^1D_2$	1522.4953	1522.0793	196.9086	-	5.030e-02	2.361e-08	33.9	forbidden line within the ground configuration
$4p^2\ ^3P_2 - 4p^2\ ^1D_2$	1749.6778	1749.2001	171.3415	-	9.800e-02	2.361e-08	66	forbidden line within the ground configuration

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁷³Ge hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$4p^2\ ^3P_1$	15.5480(18)	-54.566(9)	F=7/2: -110.52	F=9/2: +20.83	F=11/2: +56.32	
$4p^2\ ^3P_2$	-64.4270(7)	111.825(13)	F=5/2: +759.95	F=7/2: +485.53	F=9/2: +160.66	F=11/2: -193.68 F=13/2: -551.89
$4p5s\ ^3P^o_0$	-262.2(12)	40(6)	F=7/2: +1460.43	F=9/2: +235.53	F=11/2: -1169.90	

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4p^2\ ^3P_1 - 4p5s\ ^3P_1^o$	269.214	70-72	-35.0(20)
$4p^2\ ^3P_1 - 4p5s\ ^3P_1^o$	269.214	73-72	8(4)
$4p^2\ ^3P_1 - 4p5s\ ^3P_1^o$	269.214	74-72	52.0(20)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$